

```
In [1]: # A pivot table is used to summarize and analyse data by grouping rows and
#applying aggregate functions such as sum,mean,count,max,min,etc....
```

```
#step1: Load dataset
```

```
import pandas as pd
df = pd.read_csv("people-100.csv")
print(df.head())
print(df.columns)
```

	Index	User Id	First Name	Last Name	Sex	\
0	1	88F7B33d2bcf9f5	Shelby	Terrell	Male	
1	2	f90cD3E76f1A9b9	Phillip	Summers	Female	
2	3	DbeAb8CcdfeFC2c	Kristine	Travis	Male	
3	4	A31Bee3c201ef58	Yesenia	Martinez	Male	
4	5	1bA7A3dc874da3c	Lori	Todd	Male	

		Email	Phone	Date of birth	\
0		elijah57@example.net	001-084-906-7849x73518	1945-10-26	
1		bethany14@example.com	214.112.6044x4913	1910-03-24	
2		bthompson@example.com	277.609.7938	1992-07-02	
3		kaitlinkaiser@example.com	584.094.6111	2017-08-03	
4		buchananmanuel@example.net	689-207-3558x7233	1938-12-01	

	Job Title
0	Games developer
1	Phytotherapist
2	Homeopath
3	Market researcher
4	Veterinary surgeon

```
Index(['Index', 'User Id', 'First Name', 'Last Name', 'Sex', 'Email', 'Phone',
      'Date of birth', 'Job Title'],
      dtype='object')
```

```
In [2]: #step2:Create Age Column
#convert str to date
df['Date of birth'] = pd.to_datetime(df['Date of birth'])
today = pd.Timestamp.today()
df['Age'] = (
    (today - df['Date of birth']).dt.days // 365
)
print(df[['First Name', 'Sex', 'Age']].head())
```

	First Name	Sex	Age
0	Shelby	Male	80
1	Phillip	Female	116
2	Kristine	Male	33
3	Yesenia	Male	8
4	Lori	Male	87

```
In [4]: #1.Basic Pivot Table
#Average age by Gender
pivot1 = pd.pivot_table(
    df,
    values='Age',
    index='Sex'
```

```

)
print(pivot1)

#2.pivot table with aggfunc='mean',
pivot2 = pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc='sum'
)
print(pivot2)

```

	Age
Sex	
Female	64.981132
Male	54.957447

	Age
Sex	
Female	3444
Male	2583

```

In [6]: #3.pivot table with aggfunc='count',
pivot3 = pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc='count'
)
print(pivot3)

#4.multiple aggregations
pivot4 = pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc=['mean', 'max', 'min']
)
print(pivot4)

```

	Age
Sex	
Female	53
Male	47

		mean	max	min
	Age	Age	Age	Age
Sex				
Female	64.981132	117	8	
Male	54.957447	117	5	

In []:

In []: